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Event Log Correlation for Multi-Step Attack Detection

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ABSTRACT

Event log correlation (ELC) is central to detecting multi-step attacks (MSAD) that unfold across heterogeneous systems and long time horizons. This review synthesises ELC families—mining/sequence, graph learning, provenance/causal correlation, and hybrid LLM-assisted approaches—through an MSAD-first lens that ties methods to attack stages and datasets. We report operational metrics (false-alarm reduction, detection time, throughput/storage) and classifier metrics (Accuracy/F1) as the authors present them, enabling fair comparison across 2025 works. Compared with prior surveys, we contribute a challenge → mitigation map (false positives, latency/throughput, heterogeneity), a 2025-only section covering NDSS/USENIX/Neurocomputing studies and a recent Graph Convolutional Network (GCN) article (Multi dataset, Multi-family detection pipeline), and a roadmap spanning mining, graph, provenance/causal, and LLM-assisted correlation for scalable, real-time deployments. We also provide an attack-coverage matrix and a machine-readable extraction (8 papers × 15+ fields) to support reproducible synthesis and practitioner adoption.

1 | Introduction

Modern computer systems (such as IoT devices) generate vast volumes of structured and unstructured data, making threat MDI (monitoring, detection, identification) increasingly complex. Attackers exploit this complexity through multi-step attacks (MSAs), where adversaries execute coordinated or correlated events to bypass traditional security defences. Every MSA triggers unique events, and their detection requires event log correlation, which links security events from heterogeneous sources such as system logs, network traffic, and application logs and finds crossover between them to reconstruct attack chains. Compelling correlation enables security analysts to identify attack progression, detect hidden patterns, and proactively mitigate threats. An important issue: Enterprise hosts routinely emit millions of audit/endpoint detection and response (EDR) events daily, with $10^3 - 10^4$ alerts/day in busy security operations centers (SOCs); correlation is essential to suppress false positives and surface multi-step campaigns.

ELC techniques leverage statistical methods, machine learning, deep learning, and graph-based approaches to reduce data redundancy, extract meaningful patterns, and minimise false positives. Because these logs have become increasingly diverse and voluminous, scalability, noise reduction, and real-time correlation remain active research challenges. Mining-based approaches such as process mining, sequential pattern mining, anomaly detection, and clustering offer promising results for IDS, IPS and SEIM tools by uncovering implicit attack/attacker relationships and behavioural patterns within event logs. While some researchers have examined ELCs, existing reviews often lack a dedicated focus on MSAD, dataset suitability concerning

Abbreviations: ELC, event log correlation; MSADP, multi-step attack detection and automation; RAG, retrieval-augmented generation; SLR, systematic literature review.

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various scenarios, and mining-based techniques for log correlation. This review fills this gap by analysing event correlation methods for multi-step attack detection, evaluating their effectiveness in linking attack phases, improving event interpretation, and reducing false alarms. In addition, we examine publicly available and synthetic datasets, assessing their applicability to event log correlation research. Although numerous publications have addressed event log correlation, a comprehensive systematic assessment of mining-based correlation techniques, dataset appropriateness, and real-time applicability specifically for MSAD still requires attention. This review directly tackles those deficiencies.

A recent study proposes the creation of customised synthetic datasets utilising large language models (LLMs) in conjunction with Retrieval-Augmented Generation (RAG) to offer varied and authentic attack scenarios absent in traditional datasets.
Systematic analysis. We review event-log correlation (ELC) for multi-step attack detection (MSAD) across five families—rule/statistical correlation, mining/sequence models, graph learning, provenance/causal correlation, and hybrid LLM-assisted methods—showing how each family supports different MSAD stages (reconnaissance, privilege escalation, lateral movement, exfiltration). We contrast *classifier metrics* (Accuracy, Precision, Recall, F1) with *systems/streaming metrics* (false-alarm reduction, detection time, throughput, storage). We ground the discussion in the 2025 literature synthesis (Tables 3–5).

A classification of mining-based approaches. Within mining/sequence methods, we distinguish (i) *process mining* for deriving control-flow models from logs; (ii) *sequential pattern mining* for stage ordering and cross-host linkage; (iii) *anomaly detection* for rare or out-of-profile sequences; and (iv) *clustering* for grouping semantically similar traces. We note practical considerations—log parsing, feature construction, windowing, and concept drift—and summarise where mining excels (stage inference, portability) and where it is weaker (causal attribution, strict online latency) (Section 4).

Evidence-based dataset evaluation and augmentation. We evaluate commonly used datasets for MSAD realism and reproducibility—e.g., DARPA Transparent Computing, OpTC, CTU-13, CICIDS2017, UNSW-NB15—considering heterogeneity (host, network, provenance), label fidelity, temporal span, and attack coverage. We discuss gaps (e.g., long-horizon ground truth, cross-domain generalisation) and outline how synthetic generation and *LLM–RAG* augmentation can create balanced, stage-aware corpora without compromising validity (Section 3 and Table 3).

Visualisation and practitioner roadmap. To support method selection, we provide an evidence-based visual mapping that situates techniques by log source, correlation mechanism, and deployment context, and we pair it with a *challenge → mitigation* summary linking common pain points—false alarms, latency/throughput, heterogeneity—to concrete techniques (provenance causality, task-guided streaming, stage-aware sequences, LLM explanations). This serves as a practitioner-oriented roadmap toward scalable, real-time MSAD pipelines (Figure 6 and Table 9).

This review is a foundational reference for cybersecurity researchers and practitioners, providing insights into advancements in event log correlation, data-driven security analytics, and the role of AI-driven correlation techniques in multi-step attack detection.

1.1 | Contributions and Novelty

Compared with prior ELC surveys (2019–2024) that provide broad taxonomies without an MSAD-first lens or focus on a single family, this review:

1. Focuses explicitly on *MSAD-oriented event log correlation*, **aligning ELC families to MSAD stages and attack families** and bridging rule/statistical, mining/sequence, graph, provenance/causal, and hybrid/LLM approaches with operational needs;
2. Introduces an *evidence-based dataset suitability perspective* for correlation tasks (beyond "usage of datasets"), explicitly referencing **DARPA TC, OpTC, CTU-13, CICIDS2017** and discussing the role of **synthetic/LLM–RAG augmentation**;
3. Provides a *challenge → mitigation* summary that operationalises correlation choices for common ELC pain points **(false positives, latency/throughput, heterogeneity)** with concrete techniques (**provenance causality**, **task-guided streaming**, **stage-aware sequence models**, **LLM explanations**);
4. Delivers *2025 updates* covering **NDSS, USENIX, Neurocomputing** works [1, 2, 4] and our **GCN article by**[8], ensuring recency and comparative breadth;
5. Reports and contrasts *both* **classifier metrics** (Accuracy, Precision, Recall, F1) *and* **systems/streaming metrics** (false-alarm reduction, detection time, throughput, storage) to reflect how different families have been evaluated in practice; and
6. Offers a *conceptual roadmap* aligning datasets, methods, and system requirements to guide **scalable, real-time** deployments.

1.2 | Paper Structure

1. Section 2 formalises ELC for MSAD and discusses datasets.
2. Section 3 details the PRISMA-based methodology and data extraction.
3. Section 4 surveys correlation families (mining/sequence, graph, provenance/causal, and hybrid/LLM),
4. and Section 4.2 presents the 2025 additions (scope, classifier and systems metrics, attack coverage, and critical appraisal).
5. Section 5 synthesises challenges and introduces the challenge → mitigation map
6. Section 6 outlines a quick summary of challenges reported in Section 5.

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7. Section 7 outlines a roadmap and future directions.
8. Section 8 concludes with key takeaways.

2 | ELC and Mining-Based Detection Methods for MSAD

Event log correlation (ELC) plays a pivotal role in multi-step attack detection (MSAD) by linking disparate security events from heterogeneous sources such as system logs, network traffic, and application logs. Given event logs' high-dimensional, unstructured, and voluminous nature, mining-based methods offer robust solutions for uncovering attack sequences, detecting anomalies, and deriving meaningful patterns.

Mining-based detection techniques have been extensively employed to improve correlation by identifying patterns, reducing false positives, and ensuring real-time detection. Association rule mining, sequence mining, clustering, anomaly detection, and frequent pattern mining are among the most effective approaches used in event correlation frameworks. These methods help model multi-step attacks by detecting relationships between events, analysing deviations from normal behaviour, and enhancing log analysis automation.

2.1 | Mining-Based Methods for Event Log Correlation

Several mining-based techniques have been applied to ELC to improve attack reconstruction, reduce false positives, and automate correlation processes. Examples include, but are not limited to:

2.1.1 | Association Rule Mining

Association rule minings (ARMs) help identify frequently co-occurring event sequences in logs, helpful in detecting known attack chains and insider threats [9]. Beyond co-occurrence, support, confidence, and lift can prioritise prevalent and discriminative rules for MSAD stages [10, 11]. Practical pipelines often combine ARM with temporal constraints (e.g., sliding windows) and post-filtering against benign baselines to reduce spurious associations in high-volume SOC telemetry [12].

2.1.2 | Sequence Mining

Sequence mining enables the extraction of temporal relationships between log events, enabling better modelling of attack steps in MSAD [13, 14]. Standard operators include contiguity, gaps, and time lags, which help capture multi-host lateral movement patterns. Frequent/closed sequential patterns can seed stage mappers or HMM/Gated Recurrent Unit (GRU)/Transformer models that turn raw sequences into phase-aware alerts for triage [15].

2.1.3 | Clustering

Clustering helps to group log events with similar characteristics, assisting in event aggregation, pattern detection, and false positive reduction [16, 17]. Density-based methods (e.g., DBSCAN) tolerate noise and discover irregular attack "pockets," while k-means and hierarchical clustering work well for fast coarse triage. Clustering, an unsupervised learning technique, often summarises alerts and floods and empowers profile-based safelisting in event log correlation (ELC) based MSAD pipelines [18].

2.1.4 | Classification

By labelling log entries according to predetermined attack categories, classification or supervised learning increases the effectiveness of detection [19, 20]. To prevent overly optimistic accuracy/overfitting, robust configurations report per-stage data (Precision/Recall/F1) and manage class imbalance (e.g., SMOTE, focal loss). Portable feature sets (host + network) improve generalisation when schemas drift across sensors or organisations [15].

2.1.5 | Anomaly Detection

Anomaly detection identifies deviations from normal system behaviour, helping detect previously unknown attack patterns [21–23]. Unsupervised models (e.g., LOF, Isolation Forest, auto-encoders) reduce reliance on labelled data and surface long-tail behaviours typical of stealthy multi-step campaigns. Calibrating thresholds with seasonality and business-hour priors lowers false alarms in production.

2.1.6 | Frequent Pattern Mining

Frequent pattern mining (FPM) discovers frequently occurring attack patterns in large-scale event logs [24]. For MSAD, episodes (partially ordered events) and closed/maximal patterns cut redundancy and map naturally to tactictechnique progressions. In streaming data, these mined patterns frequently serve as seed rules or graph motifs that downstream correlators match [13].

2.1.7 | Outlier Detection

Unexpected log sequences that could point to new or covert attack pathways can be found with the aid of outlier detection [21, 25]. Outlier scores have the ability to identify sessions for more thorough forensic reconstruction, re-ranking of alerts or initiating context expansion (such as retrieving adjacent host/process events) in correlation procedures [17]. Combining outlier criteria at the point and sequence levels works exceptionally well for multi-phase attacks.

2.1.8 | Dimensionality Reduction

For better correlation, dimensionality reduction streamlines log data while preserving essential traits [18, 26]. Neural (autoencoder-AEs) and linear (PCA) projections reduce high-cardinality, sparse fields (such as attack paths and registry keys) to denser embeddings that enhance both supervised and

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clustering models. Mini-batch or incremental variations maintain throughput throughout the deployment process without requiring retraining.

2.1.9 | Text Mining

Text mining helps analyse free-text log descriptions for additional contextual correlation insights [27]. Topic models (e.g., LDA) and embedding models (e.g., doc2vec/transformers) map alerts to semantic clusters, improving deduplication and linking semantically similar steps across hosts [12]. Simple keyword expansions (synonyms, vendor aliases) also boost recall in noisy ticket/alert notes.

2.1.10 | Large Language Models

Using AI-based methods, large language models (LLMs) automate log processing, summarisation, and anomaly detection [28, 29]. LLMs are utilised extensively in MSAD/ELC with RAG across artefact knowledge bases, schema normalisation across heterogeneous sources, and narrative production (stage-wise explanations). Guardrails (quick discipline, cost/latency budgets, and red-teaming) are necessary for dependable SOC integration [5].

2.1.11 | Process Mining

In a study [30] where authors presented the EC-SA and EC-SA-Data (Event-Case Correlation Using Simulated Annealing) approaches for event-case correlation (ECC) in logs without explicit case IDs, they solidified the need for process mining. This approach uses categorisation, simulated annealing optimisation, and process model compliance to restore event-case linkages or crossovers. The flexibility to support different logs/log formats (diversity) is a key plus, especially for cybersecurity and forensics logs, where case IDs are sometimes missing or unclear (a condition for forensic readiness). Although the approach enhances trace creation and downstream mining accuracy, its reliance on preset models and restrictions may reduce flexibility in dynamic MSAD attack scenarios [31]. Furthermore, there is a limited direct operational generalisability due to the lack of a thorough evaluation of scalability and performance indicators in large-scale log ecosystems (such as MSAD or SIEM systems). Pre-processing of event logs is a crucial but often overlooked part of process mining for efficient MSA detection. A research study [32] categorised current methods for log filtering, abstraction, timestamp alignment, and clustering in their thorough review. This research offers the foundation for cybersecurity experts who work with noisy, heterogeneous logs, often within MSAD processes. Although, the study covers a lot of ground, it provides no empirical comparison and no real-world scenarios for security event connection. Therefore, the relationship to MSAD is relevant in the basic data cleaning and transformation techniques rather than the event correlation mechanisms. Table 1 highlights each mining/sequence strategy's primary role in correlation pipelines by outlining how each contributes to event-log correlation for MSAD.

2.2 | Dataset for Multi-Step Attack Detection

The selection of datasets is critical in evaluating multi-step attack detection (MSAD) techniques. Our analysis of the literature reveals that various datasets have been employed, with some being more frequently used than others. DARPA 2000 [33], NSL-KDD [17], and CICIDS2017 [4] are the most commonly utilised datasets, reflecting their relevance and accessibility for evaluating MSAD models. In the context of event log correlation (ELC), datasets such as DARPA 2000 [33], LLDOS1.0 [34], Mordor [33], LANL [35], and EDR Alerts have been leveraged due to their structured event logging, enabling correlation of attack steps across different log sources. These datasets facilitate the reconstruction of attack sequences and support methods that rely on graph-based modelling, sequence analysis, and alert correlation. For mining-based techniques, common datasets such as NSL-KDD [17], CICIDS2017 [4], UNSW-NB15 [17], and National Vulnerability Database (NVD) [36] are extensively used. These datasets contain labelled attack instances, enabling the use of association rule mining, clustering, anomaly detection,

TABLE 1 | Mining-based methods for event log correlation in MSAD.

Mining-based method	Role in event log correlation
Association rule mining	Identifies co-occurring attack behaviours across different logs to build correlation rules
Sequence mining	Detects time-ordered attack sequences, improving multi-step attack reconstruction
Clustering	Groups similar log events for efficient attack chain modelling and reduces redundant alerts
Classification	Categorises event logs into benign or malicious groups to facilitate automated correlation
Anomaly detection	Identifies outliers in event logs that may indicate unknown attacks or zero-day exploits.
Frequent pattern mining	Extracts recurring attack sequences to model known attack behaviours
Outlier detection	Identifies rare or stealthy attack sequences that evade signature-based detection
Dimensionality reduction	Enhances correlation efficiency by reducing irrelevant log features while preserving attack signatures
Text mining	Analyses free-text log descriptions for additional contextual correlation insights
Large language models (LLMs)	Automates log processing, summarisation, and anomaly detection using AI-based techniques

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and other data-driven methods. They provide structured and semi-structured attack logs, which are well-suited for pattern extraction and behavioural analysis. Moreover, synthetic datasets (e.g., Synthetics [37], MITRE ATT&CK [38], and CERT APT [37]) are widely adopted to bridge real-world data gaps. These datasets provide tailored attack scenarios that support benchmarking of advanced MSAD techniques. The diversity of datasets highlights the need to choose datasets that are aligned with the detection technique under evaluation to ensure robust and generalisable model performance. Figure 4 presents an overview of these datasets.

2.3 | Integrating Mining-Based Techniques for MSAD Event Log Correlation

The effectiveness of ELC depends on identifying meaningful correlations between security events while reducing false positives. Mining-based methods improve correlation effectiveness by:

i. detecting attack sequences in logs to reconstruct multi-step attack chains;

ii. improving automated alert triage to reduce manual security analyst workload;

iii. identifying novel attack techniques through anomaly detection and outlier analysis;

iv. increasing event correlation efficiency in real-time scenarios;

v. leveraging AI-driven models (e.g., LLMs) to improve log-based threat intelligence.

While traditional correlation techniques rely on predefined rules, data mining enables more dynamic, adaptive, and scalable approaches to log-based multi-step attack detection. By combining mining-based event correlation with AI-driven insights, MSAD frameworks can improve accuracy, efficiency, and attack predictability in cybersecurity environments.

3 | Research Methodology

This review adheres to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) to guarantee a systematic, transparent, and reproducible methodology for MSAD utilising event log correlation. The process encompasses (i) identification, (ii) screening, (iii) eligibility assessment, and inclusion criteria of appropriate papers according to a clearly articulated research protocol. This investigation employs PRISMA to guarantee a systematic and impartial selection of literature, assessing the most pertinent event correlation approaches, datasets, and security analytics techniques investigated in multi-step attack detection research.

3.1 | Research QuestIons

The research questions (RQs) guiding this systematic review are:

1. What primary event log correlation techniques are used for multi-step attack detection (MSAD)?
2. How do mining-based methods (e.g., process mining, sequential pattern mining, anomaly detection, clustering) contribute to event log correlation?
3. What datasets have been used for event log correlation (ELC), and how suitable are they for MSAD using evaluation metrics?
4. What evaluation and performance metrics are used to assess ELC techniques in MSAD?
5. What are the significant challenges and research gaps in existing research, and how can LLM/RAG-based synthetic datasets improve event correlation for MSAD?

These questions structure the review process to ensure that only relevant studies addressing event correlation for MSAD are included in the final analysis.

3.2 | Search Strategy and Data Sources

A systematic search was conducted across the following academic databases to ensure comprehensive coverage of relevant studies. (i) IEEE Xplore (22); (ii) Scopus (24); (iii) Google Scholar (Supplementary search).

3.2.1 | Recency Update 2025

Following peer review, we extended the search window to include studies published in 2025. The inclusion/exclusion criteria remained unchanged. The search used Boolean operators to include variations of event correlation techniques and MSAD-related terms.

The keywords and search strings were:

```
Search Direct Query: 22 Articles
(event log correlation) AND ((multi-step
attack detection) OR (multi-stage attack
detection) OR (APT detection) OR (DDos
detection) OR (Ransomware detection) OR
(Multi-phase attack detection) OR (Com-
posite Attack detection))
```

This systematic review encompassed studies from January 2000 to January 2024, utilising the IEEE Xplore, Scopus, and Google Scholar databases. Furthermore, backwards and forward citation analysis was conducted to uncover further relevant articles. Eight more relevant articles were also included from 2025.

3.3 | Inclusion and Exclusion Criteria

Studies were included or excluded to maintain focus and quality based on rigorous selection criteria.

3.3.1 | Inclusion Criteria

1. Studies published from Inception till 2024 focusing on event log correlation for MSAD.

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TABLE 2 | Evaluation criteria for event log correlation studies in MSAD.

Criteria	Question	Score (0–5)
Relevance	Does the study address event log correlation for MSAD?	0–5
Methodology	Is the research method clearly defined and validated?	0–5
Experimentation	Does the study provide experimental results on datasets?	0–5
Dataset Suitability	Does the study use real or synthetic datasets?	0–5
Significance	Does the study contribute novel insights to event correlation?	0–5

2. Research that explicitly discusses security event correlation techniques (e.g., graph-based, rule-based, ML-based, statistical, mining-based).
3. Papers evaluating security logs or synthetic datasets for ELC.
4. Studies presenting performance evaluations of ELC-MSAD.

3.3.2 | Exclusion Criteria

1. Papers unrelated to security event correlation or MSAD.
2. Articles focusing only on single-step attacks without correlation techniques.
3. Purely theoretical studies without experimental validation or dataset usage.
4. Non-peer-reviewed or low-quality sources (e.g., editorials, opinion pieces, non-English sources).

3.4 | Data Extraction and Categorisation

The final 120 selected studies were analysed and categorised based on the following criteria:

Correlation Techniques (12 Studies): Further divided into (i) graph-based techniques; (ii) machine learning-based approaches; (iii) mining-based methods; (iv) statistical and rule-based techniques.

Datasets Used in Event Log Correlation (12 Studies): (i) available datasets; (ii) custom and synthetic datasets; (iii) suitability metrics.

Performance Evaluation Metrics (12 Studies): (i) false positive rate (FPR), true positive rate (TPR), precision, recall, and F1-score; (ii) event correlation accuracy and scalability metrics.

Challenges and Future Research Directions (51 Studies): (i) No article available that uses LLM-based synthetic datasets for event log correlation; (ii) 44 articles that report ELC-based challenges; (iii) Some articles discussing causal inference techniques for MSA attribution.

3.5 | Quality Assessment Criteria

Each source was evaluated based on a Quality Assessment (QA) Score, ensuring that only high-quality papers contributed to the review. Articles were scored from 0 to 5 on the criteria discussed in the Table 2.

Studies with QA scores ≥ 3 were included in the final analysis.

3.6 | PRISMA Flow Diagram

This section shows the PRISMA flow diagram. Figure 1 shows the PRISMA diagram used in the study.

4 | Related Work

This highlights studies relevant to ELC with respect to mining-based, graph-based, and clustering-based approaches for multi-step attack detection. The first research uses a mining-based method by [10] employed data mining techniques on web server access logs, addressing the issue of class imbalance using SMOTE. Achieving F1-scores of 0.99, 0.97, and 0.96, their approach demonstrated strong anomaly detection capabilities but lacked scalability analysis. Another mining-based correlation approach by [12] used data mining techniques with Latent Dirichlet Allocation (LDA) and Random Forest for feature extraction and alert correlation, achieving 98% accuracy on SNORT alerts. Despite its promising results, the approach remains untested in real-time environments and larger datasets. Refer to Figures 2 and 3 for a visual representation of detection methods.

4.1 | MSA Detection Families at a Glance

Mining/sequence methods capture temporal regularities and support stage inference; graph learners capture structural dependencies; provenance/causal ELC enforces multi-evidence attack paths and optimises (False Alarms (FA)/latency/throughput); hybrid/LLM methods enhance explanations and analyst workflows. We position these as complements, not competitors, and map each to MSAD challenges.

A method using graph mining was employed for event correlation, focusing on multi-step attack detection with the DARPA dataset [13]. Their approach effectively modelled causal relationships but did not validate scalability or applicability to broader datasets. Some graph-based research studies include research by [39] using explainable graph neural network (GNNs) to analyse malware traffic data, achieving 99% accuracy on the Malware Capture CTU-13 dataset. This method offers interpretability and robustness but lacks validation on dynamic and heterogeneous

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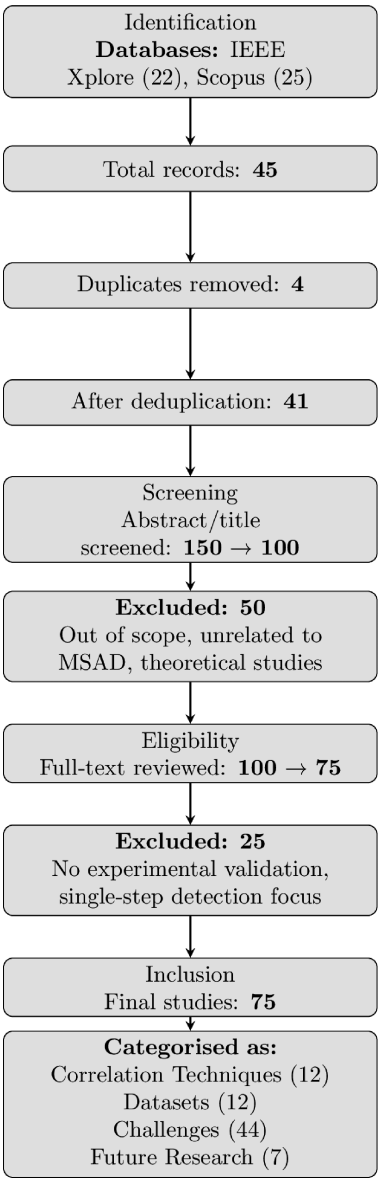

FIGURE 1 | PRISMA flow diagram showing the selection process.

datasets. Temporal-Gated (GNNs) combined with GRU system for MSAD was proposed by [40], achieving 95.14% accuracy on UNSW-NB15 datasets. Their work demonstrated the power of temporal correlations in improving detection accuracy, though the high computational overhead limits its practical applicability for real-time environments. The contribution by [33] used temporal-topological correlation to structure alert relationships and reduce false positives. Their work integrated graph mining, offering a robust method for temporal correlation. However, the reliance on DARPA datasets restricts its application to broader real-world scenarios.

Building on iterative updates for event correlation, Liao et al. [37] proposes a dynamic risk assessment framework. The framework integrates IDS setups, Snort rules, and synthetic datasets that model attacks such as EternalBlue and Heartbleed. While computationally demanding and reliant on simulated datasets, this structured approach organises attack-defence knowledge using a Status-based Attack Tree (S-AT) but limits generalisation across diverse attack paths. Navarro-Lara et al. [44] created a hybrid correlation approach and applied reinforcement learning to reduce false positives in alert correlation. Their method processed 40 000 events in 1836 ms, demonstrating efficiency in heterogeneous SIEM logs. However, the absence of detailed accuracy metrics and scalability testing limits the generalisability of the approach.

The study by [45] proposed a correlation and aggregation framework that significantly reduced false positives in DARPA and Insider datasets. However, despite integrating graph-based methods, the framework lacked execution time analysis and validation on diverse datasets. Another study by [46] combined Bayesian attack subgraphs with a probabilistic autocorrelation method to detect attack patterns. Their approach achieved high TDVC values across clustering algorithms, demonstrating its effectiveness in pattern detection. However, manual vectorisation and limited scalability testing pose challenges in automation and scalability. Another contribution by [33] used temporal-topological correlation to structure alert relationships and reduce false positives. Their work integrated graph mining, offering a robust method for temporal correlation. However, the reliance on DARPA datasets restricts its application to broader real-world scenarios.

An important contribution by [47] developed a state machine correlation method for the causal inference-based detection method. Their approach, tested on DARPA datasets, effectively modelled sequential dependencies but failed to validate scalability or real-time performance. Data fusion-based research (A Multi-Modal Event Correlation) by [48] focused on information fusion using ontology to address data heterogeneity across datasets like ATT&CK and CAPEC. This approach demonstrated potential for improving multi-domain correlation but lacked empirical performance metrics and real-time validation.

A scalable and effective approach, proposed by [49], employed graph-based correlation and aggregation to process alerts, focusing on identifying causal links. Their method, tested on DARPA datasets, highlighted the value of structured correlation but did not include performance evaluation or scalability analysis. Similarly, Yuen et al. [50] explored attack graph visualisation for correlating attack paths. While the visualisation provided insights into vulnerabilities, the absence of performance evaluation and real-world validation limited its applicability. A factor graph-based approach by [14] for temporal-topological correlation, achieving a 96% reduction in false positives on the DARPA 2000 dataset. This research can be classed as a causal inference-based correlation technique using temporal correlations and enhanced alert correlation, but scalability and real-time performance remain unexplored. These studies collectively highlight the potential of correlation techniques in identifying multi-step attacks and reducing false positives, but underscore the need for real-time validation and scalability improvements. Another article by [51] in a DL-based correlation approach, which employed DenseNet CNNs to improve Advanced Persistent Threat (APT) detection, significantly improved classification accuracy. However, the work does not address computational overhead or performance in high-traffic environments. In another category, [41] (Machine Learning for Event Fingerprinting in APTs) used a hybrid approach with PCA for feature reduction and anomaly detection, achieving 97.63% accuracy on the APT 2020 dataset. While

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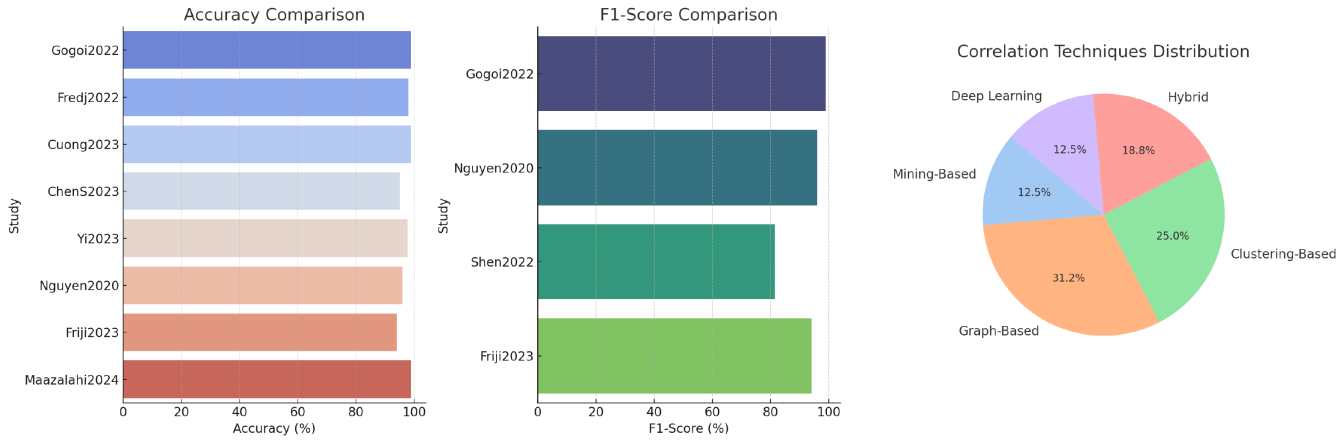

FIGURE 2 | Visual representation of event log correlation methods, illustrating their suitability for different multi-step attack detection scenarios. Due to limitations of graphic design tools, while writing, we had to use a different citation style in the figure. For quick reference: Gogoi2022 [10], Fredj2022 [12], Cuong2023 [39], ChenS2023 [40], Yi2023 [41], Nguyen2020 [42], Friji2023 [22], Maazalahi2024 [17], Shen2022 [43]. Figure 3 aligns this figure with detection methods with sources. For more insights, please read section: related work Section 4.

Family	Role in MSAD (example references from this review)
Mining / sequence	Captures temporal regularities; supports stage inference and cross-host linkage; helps triage. Citations: [3,23,8,13,12,10,41,6,7,11]
Graph learning (GNNs)	Models structural dependencies and long-range relations; sometimes offers explainability. Citations: [36,37,42,43,44,18]
Provenance / causal ELC	Enforces multi-evidence attack paths; reduces false alarms, and supports online segmentation/streaming. Citations: [1,2,45,46,32,47,48]
Hybrid LLM-assisted	LLM narratives / RAG augmentation for explanations, dataset synthesis, and analyst workflow. Citations: [26]
SOC alert contextualisation	Alert grouping, historical matching, and triage; complements MSAD correlation. Citations: [49,50]

FIGURE 3 | Visual mapping of event-log correlation families to MSAD scenarios. *Please note.* Detection methods are complementing: mining/sequence (stage semantics), graph learning (structure), provenance/causal (multi-evidence), hybrid LLM (explanations/augmentation), and SOC contextualisation (triage).

effective in handling high-dimensional data, its dependence on dataset-specific parameters restricts its generalisability.

Some contributions regarding clustering and hybrid detection methods for MSAD include the following: The author of [42] combined PSO with Local Outlier Factor (LOF) to optimise clustering parameters, achieving high anomaly detection accuracy. However, the computational overhead of PSO remains a challenge for real-time applications. Another contribution where [43] employed NSGA-2 for multi-objective optimisation in clustering and classification, achieving an F1-score of 81.37% on the SCVIC-APT-2021 dataset. While their approach optimised clustering, the moderate F1-score and lack of real-world testing highlight gaps in its application. This research by [52] integrated multi-objective PSO with Support Vector Machines (SVM), achieving near-perfect accuracy on IoT botnet datasets. Despite its success, the lack of validation in real-time and dynamic IoT environments limits its broader applicability. Another research by [22] utilised Graph Neural Networks (GNNs) combined with Random Forest and RNN to detect anomalies in IoT datasets. Their method achieved an F1-score of 94% on the ToN IoT dataset, with clustering aiding in preprocessing. While the graph-based approach is effective in capturing relationships between events, its scalability and generalisability to non-IoT datasets remain untested.

In [53], the authors applied Hidden Markov Model (HMM) for network correlation, achieving efficient performance across HTTP and SNORT datasets. However, inconsistent execution times and reliance on specific datasets reduce its adaptability to diverse scenarios. An approach proposed by [36] incorporated Ant Colony Optimization (ACO) for vulnerability categorisation, achieving fast processing times. Although ACO showed promise, its reliance on vulnerability-specific datasets limits its generalisability. A novel approach by [17] employed PSO, Genetic Algorithm (GA), and Antlion Optimisation (ALO) for feature selection and anomaly detection. Their method achieved high accuracy across datasets like NSL-KDD, UNSW NB15, and KDD CUP99. While these techniques improved performance, the computational cost and lack of scalability testing remain critical limitations. Figure 4 shows various datasets used in literature for MSAD and their suitability to Multi-Step Attacks (MSAs), Large Language Model (LLMs), Synthetic Data Suitability (SDS), Event Log Correlation (ELC), High Dimensionality (HD), False Positive

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Dataset	MSA	LLM	SDS	ELC	HD	FPMD	PPC	Data Structure & Volume	Attack Covered	Annotation Quality & Labels	Dist	Challenges & Limitations
	Suitability											
DARPA 2000								Moderate - 5GB, network logs	DDoS, R2L, U2R, Probe	Binary labels	P	Outdated attacks, high false positives
NSL-KDD								Small - 150K samples	R2L, U2R, DoS, Probe	Binary, reduced redundancy	P	Synthetic, lacks real-world diversity
CICIDS2017								Large - 3M records, PCAP, CSV	DDoS, Botnet, Brute Force	Detailed labels, timestamps	P	Imbalanced attack representation
UNSW-NB15								Moderate - 2M samples, mixed	DoS, Fuzzers, Reconnaissance	Multi-class, network features	P	Requires extensive preprocessing
CTU-13								Moderate - 13 botnet captures	Botnet	Binary labels, timestamps	P	Limited to the botnet, lacks attack variety
LLDOS1.0								Small - 15 MB	DDoS	Minimal labels	P	Simple attacks, not diverse enough
CICDDoS2019								Large - 50GB	DDoS	Multi-label, extensive timestamps	P	DDoS-focused, requires high storage capacity
Mordor								Large - 500 GB Windows logs	Varied (APT-like)	Timestamped, endpoint telemetry	P	High dimensionality, complex preprocessing
DEFCON 11								Variable - varies per CTF event	Capture-the-flag	Limited labelling	P	CTF-specific, challenging for real-world use
LANL								Very Large - 1TB logs	Insider threat	High-quality, time stamped logs	R	Massive storage, complex labelling scheme
VirusShare								Large - 5TB of malware samples	Malware	Binary executables	R	Limited use for log correlation
AWID								Moderate - WiFi traces	WiFi intrusion	Binary labels, simple attributes	P	Limited application outside WiFi attacks
NVD								Moderate - Vulnerability data	Vulnerabilities	Detailed, CVE-based	P	Not suitable for live threat detection
APT3x								Variable	APT	Multi-label, targeted events	P	Limited to specific attack sequences
Ton IoT								Moderate - IoT-specific logs	IoT, network, system	Timestamped logs, detailed labels	P	Needs sophisticated parsing for event correlation
Mitre Att&ck								Extensive (broad coverage)	Broad (TTPs)	Multi-label with tactic IDs	P	Needs custom pre-processing and filtering
ISCX IDS 2012								Moderate - 2GB network logs	DoS, R2L, U2R	Binary labels	P	Outdated, lower diversity in modern attacks
WUSTL IIoT 2021								Large - IIoT logs	IIoT, network intrusions	Timestamped, multi-label	P	Requires complex preprocessing for real-world setups

FIGURE 4 | Rotated table for event log datasets with "Suitability" below column headers. Red = Low, Amber = Moderate, Green = High, Bright Green = Very High, Purple = Variable. Abbreviations: MSA: Multi-Step Attacks, LLM: Large Language Model, SDS: Synthetic Data Suitability, ELC: Event Log Correlation, HD: High Dimensionality, FPMD: False Positive Detection and Management, PPC: Post-Processing Complexity. Dist means distribution: P = Public, R = Restricted.

Detection and Management (FPDM), and Post-Processing Complexity (PPC), etc.

Graph-based techniques and mining methods, especially clustering and anomaly detection, exhibited the greatest potential for practical application owing to their scalability and precision.

4.2 | 2025 Additions (Event Log Correlation-ELC for MSAD)

Table 3 summarises eight 2025 studies spanning provenance/causal (CAS), graph (GPH), mining/sequence (MIN), hybrid LLM-assisted (HYB), and one SLR. We then link *classifier-style* results (Table 4), *systems/streaming* results (Table 5), *attack coverage* (Table 6), and a concise *critical appraisal* (Table 7).

4.2.1 | Case A: Enterprise APT (LM/PE)

On DARPA TC streams, provenance-centric ELC [1–3] correlates process, file, and socket events into attack paths, cutting false alerts (>90% in [1]) and enabling near-real-time flagging during lateral movement and privilege escalation, with resource-aware operation (Table 5).

4.2.2 | Case B: Campus DDoS/Botnet Triage

On CTU-13/CICIDS2017 and live SOC alerts, stage-aware sequences [4], alert-graph contextualisation, and our phased GCN correlate heterogeneous evidence, provide multi-MSA coverage, and reduce analyst load (Tables 4 and 6).

4.2.3 | Operational Constraints in Real-Time SOCs

Real-world MSAD detection faces (i) throughput/back-pressure under bursty telemetry, (ii) end-to-end latency budgets for triage, (iii) storage costs of long-horizon provenance, and (iv) schema drift across log sources. The 2025 systems [1–3] explicitly report FA reduction, detection time, storage/throughput, and resource use; sequence/graph/LLM works complement stage semantics and multi-MSA coverage. We therefore report both metric families and discuss deployment trade-offs.

4.2.4 | Analytical Point

Multi-evidence provenance causality reduces spurious correlations because the probability of independent benign events jointly satisfying a causal path decays multiplicatively with path length; adaptive thresholds further suppress false alerts. Task-guided segmentation/backbones compress graphs

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TABLE 3 | 2025 additions—scope at a glance.

Ref.	Venue	ELC	MSAD focus	Datasets	Log/alerts	Multi-MSA	Headline metric
[1]	NDSS	CAS	APT	DARPA Transparent Computing	Provenance (process, file, socket)	No	False-alarm reduction
[2]	USENIX Security	CAS	APT	DARPA Transparent Computing	Provenance (process, file, network)	No	Accuracy, latency, resources
[3]	USENIX Security	CAS	APT (online)	DARPA Transparent Computing, OpTC	Provenance (process backbone)	No	Accuracy, storage, throughput
[4]	Neurocomputing	MIN	Stage mapping	DARPA2000, ISCXIDS2012, CICIDS2017	IDS alerts (embedded)	Yes	Accuracy, Precision, Recall, F1
[5]	arXiv	HYB	APT (explainable)	DARPA E3, Public Arena	Provenance, LLM narratives	No	Precision, Recall, FP reduction
[6]	arXiv	GPH	SOC contextualisation	SOC alert streams	Alert graphs (time-windowed)	Yes	Matching, triage gains
[7]	JISA	—	Tools SLR	—	Mixed (network, host)	—	Quality attributes
[8]	Preprint	GPH	Multi-MSA	CTU-13, CICIDS2017, custom DDoS	Heterogeneous logs (Sysmon, EVTX, firewall, registry, PerfMon), network flows	Yes	Accuracy, F1, resources

Note: ELC families: CAS = causal/provenance, GPH = graph, MIN = mining/sequence, HYB = hybrid (e.g., LLM-assisted).

TABLE 4 | Classifier-style metrics on public datasets (as reported).

Ref.	Dataset	Metrics reported	Accuracy	F1	Notes
[4]	DARPA2000, ISCXIDS2012, CICIDS2017	Accuracy, Precision, Recall, F1	$\gtrsim 0.97$	(per-table)	Transformer stage mapping from alert sequences
[8]	CTU-13, CICIDS2017, custom DDoS	Accuracy, F1	0.9989	0.999	Heterogeneous ELC to GCN, phase-wise evaluation, resource table reported
[5]	DARPA E3, Public Arena	Precision, Recall	—	—	CADETS sequence: 25 TPs, 0 FPs, stage narratives

Note: Values are as reported by authors; some works emphasise case-level counts over aggregate F1/PR-AUC.

TABLE 5 | Systems/streaming metrics for provenance-based online detection (as reported).

References	False-alarm reduction	Latency/detection time	Throughput/storage	Notes
[1]	> 90% on avg., up to 276×	No buffer, latency near MORSE	CPU and memory reported	Adaptive gradient-tuned rule parameters over provenance
[3]	—	~ 12.78 s per GB	$> 10^4$ KB/s, up to 1806× storage reduction	Online task-guided segmentation, process backbone
[2]	—	Lower vs SOTA (tables)	Resource-efficient baseline	Simple NN exceeds complex GNNs on 8/9 DARPA datasets
[6]	—	—	—	SOC alert grouping and historical matching, triage improvements

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TABLE 6 | Attack coverage and multi-MSA capability (2025 studies).

Ref.	APT	DDoS/DoS	Botnet	Insider	Cloud LM/PE	Multi-MSA
[1]	✓					
[2]	✓					
[3]	✓					
[4]	✓	✓	✓			✓
[5]	✓					
[6]	✓	✓	✓			✓
[7]						
[8]	✓	✓	✓	✓		✓

TABLE 7 | Critical appraisal of 2025 studies (ELC/MSAD lens).

References	Pros (ELC/MSAD)	Cons/limitations
[1]	Causal/provenance correlation, large FA reductions, streaming-friendly reporting	Requires provenance layer, benign-only training assumption, fewer standard PR/F1 aggregates
[2]	Rigorous head-to-head, resource-efficient strong baseline	Single-modality provenance focus, no explicit stage labels
[3]	Online segmentation, strong throughput and storage cuts	No explicit stage tagging, relies on task segmentation heuristics
[4]	Stage-aware mapping with strong Accuracy, Precision, Recall, F1 on public sets	Single-source IDS alerts, older datasets, no causal provenance
[5]	LLM explanations reduce analyst load, low FP in case studies	Preprint status, LLM reliability/cost considerations, prompt sensitivity
[6]	Practical SOC triage improvements via grouping and history	Time-window sensitivity, metadata dependence, preprint status
[7]	Clear quality model, highlights spuriousness/missingness gaps	Not a detector, limited MSAD stage granularity
[8]	Heterogeneous ELC → GCN, **multi-family MSAD**, **multi-dataset** evaluation, high Accuracy/F1, phase-wise analysis	**Preprint status**; **standardised runtime metrics (e.g., p95 latency, events/s, AVR%) not yet included**; custom DDoS dataset documented

while preserving long-range dependencies, improving storage/latency without materially degrading sensitivity. In contrast, sequence/graph models optimised for Accuracy/F1 excel at stage semantics on public datasets but often omit strict online latency/throughput reporting. Our split presentation of classifier vs. systems metrics reflects these methodological emphases (Tables 4 and 5).

5 | Challenges in Event Log Correlation for Multi-Step Attack Detection

Event log correlation plays a critical role in identifying multi-step attacks by linking patterns across heterogeneous security logs. However, several challenges persist in effectively correlating events for Multi-Step Attack Detection (MSAD). These challenges primarily fall into five categories: data acquisition and management, event correlation complexity, log modelling and prediction, system performance, and adversarial evasion techniques. Figure 5 shows some useful insights about ELC-MSA specific challenges and their groups.

5.1 | Data Acquisition and Management

A critical challenge in MSAD is the acquisition and management of reliable, representative datasets that encompass the diversity, temporal structure, and heterogeneity of real-world attack behaviour. Most benchmark datasets (e.g., DARPA, CICIDS) are limited in scope, lack the necessary depth for causal inference, and often fail to capture multi-step attack sequences in heterogeneous environments. A recent line of research recommends not only the creation of customised synthetic datasets using large language models (LLMs) with retrieval-augmented generation (RAG) but also training with more diverse domain-specific data, which providing more realistic and variable attack scenarios, helping address these shortcomings [54, 55]. Event Log Correlation (ELC) itself is affected by the high dimensionality of event logs, where numerous attributes contribute to increased computational overhead and noise. Without effective feature engineering or dimensionality reduction, correlation algorithms struggle with scalability and efficiency [23, 56–57]. The situation becomes complicated because of the evolution of adversarial manipulation tactics, such as malicious log/code injection, modification,

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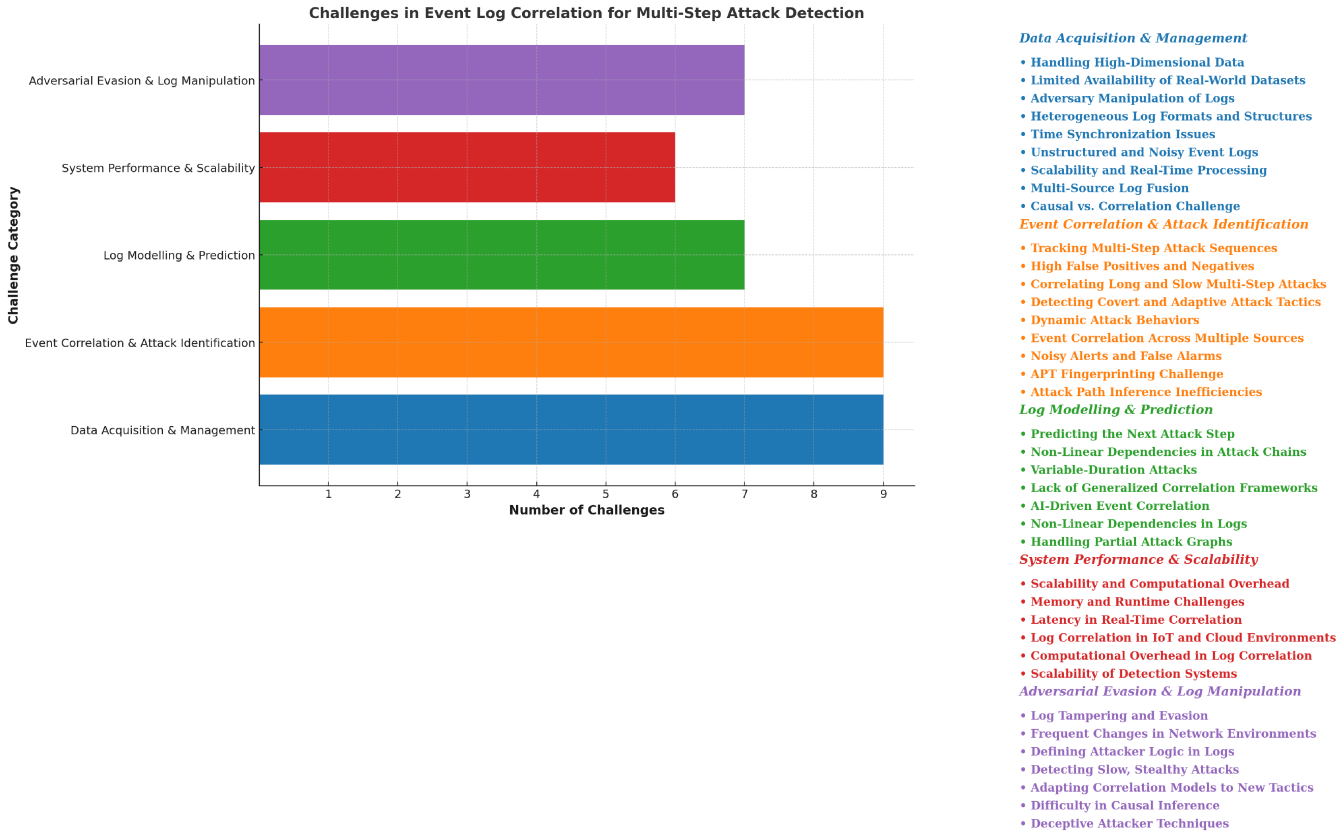

FIGURE 5 | Categorisation of MSAD event log correlation challenges and corresponding mining-based solutions, aiding practical decision-making.

or deletion, which obscure the threat detection process and compromise log integrity [43, 58]. Log heterogeneity is another major impediment. Logs from various/diverse sources (network, system, IDS, firewall) often exhibit inconsistent formats, missing fields, or semantic mismatches requiring unification, fusion or aggregation, etc. These structural disparities make standardisation and unified correlation increasingly difficult. Likewise, event synchronisation remains a concern, as timestamp discrepancies between systems operating in different time zones hinder accurate sequence reconstruction [35, 59–60]. Unstructured log content adds further complexity: free-text descriptions, redundant entries, or incomplete fields require robust Natural Language Processing (NLP) pipelines and filtering techniques to extract meaningful patterns [60, 61]. In high-volume environments, such as smart cities or cloud-based systems, real-time correlation demands optimised and parallelisable processing frameworks to avoid latency and ensure responsive detection [55, 62–63]. Finally, distinguishing causation from mere correlation remains a fundamental gap, as many existing models fail to establish causal chains between events—limiting their ability to accurately predict or respond to attack phases [54, 64].

5.2 | Event Correlation and Attack Identification

5.2.1 | Identify/Track Multi-Step Attack Sequences

Individual MSA-attack stages frequently seem normal in "isolated events" and may not activate alerts, hindering the reconstruction of comprehensive attack chains. This requires effective correlation methods incorporating temporal and semantic context to discern event sequences [65].

5.2.2 | High False Positives and Negatives

Correlation models frequently generate excessive false positives, overwhelming security teams [10, 14, 66].

5.2.3 | Correlating Long and Slow Multi-Step Attacks

Low-frequency, covert, and slow-moving attack sequences are challenging to reconstruct from event logs [25, 67–69].

5.2.4 | Detecting Covert and Adaptive Attack Tactics

Attackers frequently change their techniques to avoid detection, requiring real-time adaptation in event correlation [41, 70].

5.2.5 | Dynamic Attack Behaviours

Predefined correlation rules fail against adversaries who continuously modify their tactics [48, 69, 71].

5.2.6 | Identifying Covert/Stealthy and Adaptive Strategies in MSAs

Adversaries consistently adapt their attack methodologies to circumvent static correlation restrictions, frequently integrating

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malicious actions with legitimate communications. This necessitates adaptive, learning-oriented/intelligent correlation systems that continuously update their detection algorithms in real-time [41, 70].

5.2.7 | Noisy or Erroneous Alerts

SIEM tools frequently produce excessive alerts, many of which are unneeded or insignificant. The constant flood-like data stream renders manual correlation impractical, necessitating prioritisation tactics and automated alarm triage systems (ATS) [11, 18].

5.2.8 | MSA Fingerprinting

Generating unique attack fingerprints from logs for attacks such as Advanced Persistent Threats (APTs) is difficult due to their stealthy nature [41].

5.2.9 | Attack Path Inference Inefficiencies

Attack graph-based approaches fail to accurately infer true attack paths due to adversarial deception techniques [72, 73].

5.3 | Log Modelling and Prediction

5.3.1 | Predicting the Next Attack Step

Accurately forecasting adversarial moves based on historical logs remains unreliable due to the evolving nature of attacker strategies, de-centralised logs, and limited contextual visibility. ML-based predictors often struggle with generalising beyond training data [74, 75].

5.3.2 | Presence of Non-Linear Dependencies in MSA Sequences

Multi-step attacks do not consistently adhere to sequential or predictable patterns. The non-linear dependencies impede correlation models that presume linear progressions across attack stages [71, 76].

5.3.3 | Variable Attack Duration

MSAs such as ransomware and botnets may transpire over hours or even weeks, resulting in discrepancies in anomaly scoring and challenges in sustaining event log correlation (ELC) over time. Conventional "time-window-based" detection techniques frequently miss or incorrectly identify prolonged attack phases [71, 76].

5.3.4 | Lack of Generalisation in Correlation Frameworks

Numerous current correlation models necessitate manual rule formulation or the use of log-specific parameter adjustments. This constrains scalability across domains and obstructs real-world implementation where automation and interoperability are crucial [55].

5.3.5 | AI-Enhanced Event Log Correlation

Despite the potential for automation offered by ML and DL models, they frequently necessitate substantial pre-training and fine-tuning and exhibit difficulties in adapting to heterogeneous or novel or complex log types. This diminishes their generalisation capability in varied MSAD contexts [15].

5.3.6 | Non-Linear Dependencies in Log Files

Event sequences in MSAs may encompass concurrent activities, concealed triggers, or indirect causation, complicating correlation models that anticipate explicit temporal or logical order. This requires hybrid methodologies that integrate statistical inference with deep learning [21, 76].

5.3.7 | Management of Incomplete Attack Graphs

Attack graphs utilised to represent attacker pathways frequently exhibit incompleteness or obsolescence due to threats' dynamic natures. This increases false negatives or erroneous path predictions, particularly when encountering novel or infrequent attack vectors [73].

5.4 | System Performance and Scalability Constraints

5.4.1 | Scalability and Computational Overhead in Log Correlation

Correlating high-dimensional event logs, especially in distributed settings, imposes significant computational costs. As dataset volume and event complexity grow, traditional clustering or pattern-matching approaches suffer from processing delays and scalability bottlenecks [17, 53, 62].

5.4.2 | Memory and Runtime

Analysing logs that span terabytes of data causes memory saturation and long runtime. Some deep learning-based correlation models require hours of training or batch reprocessing, making them unsuitable for responsive or on-demand MSAD applications [40].

5.4.3 | Latency in Real-Time Correlation

Many existing correlation systems are batch-oriented or introduce latency, preventing timely identification of unfolding multi-step attacks. Low latency in ELC frameworks, including stream-based log analytics and real-time event filtering, is still underutilised in operational environments [20, 77–78].

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5.4.4 | Log Correlation in IoT and Cloud Environments

In resource-limited settings such as IoT devices and containerised cloud platforms, computing resources are often restricted. Robust ELC-focused detection algorithms cannot be implemented efficiently without offloading, compression, or approximation methods [57, 79–80].

5.4.5 | Scalability of Detection Systems

Although, major detection techniques excel in controlled settings, they do not effectively scale or perform well in real-time, high-volume enterprise contexts. The absence of horizontal scalability and restricted support for parallel computation obstructs their incorporation into production-level MSAD systems [66, 77].

5.5 | Adversarial Evasion and Log Manipulation

5.5.1 | Tampering and Evasion of Logs

Adversaries frequently alter or delete logs to obscure their actions (clear tracks), compromising the integrity of the event log timeline and obstructing accurate reconstruction of the attack event sequence. This interference hampers forensic investigations and diminishes the reliability of correlation methods [72, 73].

5.5.2 | Frequent Changes in Network Environments

Event correlation methods must adjust to evolving network topologies and configurations. Changes in protocols, device functions, or traffic patterns may cause static rules or models to become useless unless they are perpetually retrained or updated [56, 64].

5.5.3 | Defining Attacker's Logic in Logs

Advanced adversary tactics often include the injection of misleading logics into their attack sequences, utilising decoys, normal/benign events, or multi-faceted approaches to obfuscate behaviour-based models. This obfuscation complicates the extraction of significant behavioural fingerprints [48, 55].

5.5.4 | Detection of Gradual/Slow and Stealthy MSAs

MSAs are often ongoing, evolving, persistent or may evolve gradually over days or weeks. These sluggish/deceptive attacking campaigns circumvent threshold-based and "time-window-dependent" models, requiring methodologies that preserve long-term context and manage asynchronous event sequences [48, 65].

5.5.5 | Adapting Correlation Models to Novel Defence Strategies

As attack techniques rapidly evolve, correlation systems must be dynamic and resilient to zero-day patterns or unobserved behaviours. Models devoid of online learning capabilities or modularity face challenges in maintaining competitiveness [25, 41].

5.5.6 | Challenges in Causal Inference Regarding Correlated Events

Most correlation models emphasise statistical connection rather than causal relationship. This constrains their prediction efficacy, particularly in identifying sequential attack phases where one sequence or event facilitates or activates another. Novel solutions encompass causal threat modelling and graph-based inference to address this disparity [54].

5.5.7 | Deceptive Strategies Used by Attackers

Adversaries are increasingly employing deceptive/stealthy strategies, including the introduction of decoy events, the amalgamation of attack and legitimate traffic, and the imitation of regular user behaviour to circumvent detection and obfuscate correlation engines [55].

5.6 | Mapping Mining-Based Methods to Event Log Correlation Challenges

The following Table 8 illustrate the role of mining-based methods in ELC and how they address common challenges.

In addition to the mining-centric mapping in Table 8, Table 9 encapsulates pragmatic mitigation strategies—encompassing mining, graph/causal inference, and systems engineering, addressing prevalent ELC issues in MSAD.

6 | Summary of ELC Challenges in MSAD

Event log correlation for multi-step attack detection (MSAD) presents several key challenges; comprehending these challenges

TABLE 8 | Mapping mining-based methods to ELC challenges.

Challenges in ELC	Mining-based method addressing the challenge
High-dimensionality	Dimensionality reduction, clustering
Noisy and redundant logs	Clustering, outlier detection
Complex attack patterns	Sequence mining, association rule mining
Real-time correlation needs	Anomaly detection, LLMs, Generative AI
False positives in alerts	Classification, clustering, frequent pattern mining
Unknown or zero-day attacks	Anomaly detection, outlier detection
Scalability constraints	Dimensionality reduction, Generative AI

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TABLE 9 | Summary mapping of ELC challenges in MSAD to mitigation strategies and indicative techniques.

Challenge	Mitigation strategy	Indicative techniques
High dimensionality, noisy/ heterogeneous logs	Feature curation; dimensionality reduction; schema unification	Clustering/local outlier factor (LOF), representation learning, principal component analysis (PCA)/auto encoder (AE); unified parsing & typing; schema-on-read
Time sync & sequence reconstruction	Clock normalisation; temporal windows; sequence models	Network time protocol (NTP) alignment, cross-source time offset learning; seq. mining; temporal GNNs/hidden Markov model (HMMs)
False positives/alert floods	Multi-evidence correlation; risk scoring; feedback loops	Frequent pattern mining; multi-modal fusion; triage with active learning; analyst-in-the-loop
Slow/stealthy MSAs	Long-horizon state tracking; causal linkage, memory	Causal graphs/Do-calculus proxies; factor/fault graphs; hierarchical GNNs with memory
Adversarial log tampering/deletion	Integrity safeguards; provenance; redundancy	Forward-secure logging; system provenance (EDR); cross-log attestation and parity checks
Scalability/latency	Stream processing; distributed execution; approximations	Micro-batching; GPU/NUMA pinning; sketching/sampling; incremental mining
Generalisation across domains	Modular correlation layers; domain adaptation	Parameter-efficient fine-tuning; weak supervision; rule–ML hybrids with pluggable adapters
Dataset suitability gaps	Synthetic augmentation; coverage matrices; benchmarks	LLM+RAG scenario synthesis; dataset–task fit scoring; unified benchmark protocols
Attribution vs correlation (causation)	Causal inference integration	Temporal–topological causal graphs; interventional tests; counterfactual ranking

enables cybersecurity professionals to use customised correlation strategies for particular operational contexts, hence enhancing real-time threat identification and resource management in practical settings:

- **High-dimensionality:** Logs contain large, complex, multi-source data that is hard to analyse.
- **False positives and false negatives:** Correlation errors lead to alert fatigue in security operations.
- **Scalability issues:** Real-time event log correlation struggles under heavy log volumes.
- **Adversarial evasion tactics:** Attackers modify logs to obscure correlation patterns.
- **Lack of suitable datasets:** There is a shortage of real-world event log datasets for benchmarking.

Addressing these challenges requires a combination of graph-based models, machine learning, deep learning, and mining-based techniques to improve event log correlation in multi-step attack detection.

7 | Future Directions

Future research on event log correlation for multi-step attack detection (MSAD) should emphasise the development of uniform benchmarks and evaluation systems. Contemporary research depends on datasets characterised by variable scope and restricted realism, complicating comparative analysis across methodologies. Creating standardised benchmarks with rigorous real-time performance metrics, including latency, throughput, and scalability, will facilitate equitable evaluations and expedite implementation in practical settings. Figure 6 aligns dataset curation/augmentation, correlation methods (mining ↔ graph/causal), and systems concerns (latency/scale) into an actionable timeline.

A further potential approach is to incorporate causal inference into event log correlation. Although numerous methods identify the relationships or crossovers, they frequently do not distinguish between correlation and causation. Hybrid methodologies that integrate causal modelling with graph-based and mining techniques may enhance attack attribution and augment predictive capabilities, thus facilitating more effective mitigation strategies.

The utilisation of LLMs, especially in conjunction with Retrieval-Augmented Generation (RAG) to train detection frameworks, is increasingly seen as a vital area of research. LLM-driven augmentation facilitates the creation of realistic synthetic datasets replicating various adversarial strategies, infrequent attack sequences, and chaotic log environments. Consequently, incorporating synthetic data into Unified MSAD pipelines to tackle the shortcomings of real-world logs would improve model robustness and enable cross-domain testing and validation.

Scalability remains a significant issue. Enterprises, IoT, and cloud infrastructures produce gigabytes of log data each day, necessitating distributed and parallel processing methodologies. Future intrusion detection or prevention systems should explore cloud-native stream processing, GPU-accelerated analytics,

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FIGURE 6 | Conceptual roadmap for MSAD-focused event log correlation across *Datasets*, *Methods*, and *Systems*. Each track highlights near-term pragmatics, mid-term consolidation, and long-term maturity targets (benchmarks, causal inference, scalable real-time operation).

and approximate lightweight correlation techniques to enable real-time detection.

There is a growing demand for integrated frameworks capable of identifying diverse attack categories, such as APTs, Distributed Denial of Service (DDoS), botnets, ransomware, and insider threats, employing both proprietary experimental datasets and recognised benchmarks like CERT Insider Threat, CTU-13, and CICIDS2017. Under these circumstances, incorporating graph-based correlation with LLM-driven analytics highlights the potential for adaptable, multi-family detection systems that generalise across many scenarios. The advancements in these areas of research will reconcile the gap between theoretical advancements and actual applications in cybersecurity.

8 | Conclusions

Key takeaways: (i) Provenance/causal ELC [1, 3, 5] report substantial false-alarm reduction and operational efficiency on DARPA/OpTC streams; (ii) sequence/graph models achieve high accuracy/F1 on public datasets and provide stage semantics; (iii) hybrid LLM adds interpretable narratives; (iv) multi-MSA coverage is demonstrated by [4], alert-graph contextualisation, and our phased GCN; (v) future work should unify systems and classifier reporting and benchmark heterogeneous pipelines end-to-end.

This review has also illustrated the pivotal importance of event log correlation in tackling the intricacies of MSAD. Through a thorough analysis of detection methods such as graph-based, mining-based, machine learning, and deep learning methodologies, we have demonstrated how ELC facilitates the reconstruction of attack chains, enhances event interpretation, and diminishes false positives (FPs) in security operations. Techniques based on mining, such as association rule mining (ARM), clustering, and anomaly detection (AND), proved very effective for revealing links across diverse and high-dimensional/heterogeneous event log data.

The analysis of datasets indicated ongoing deficiencies or limitations in diversity, temporal breadth, and heterogeneity. Although benchmark datasets with ground truth (e.g., DARPA 2000, CICIDS2017, UNSW-NB15) are widely used, they do not fully capture heterogeneous, long-horizon MSAD scenarios encountered in practice.

Author Contributions

This review is part of a systematic literature review (SLR), which builds the foundations of the PhD research of Syed Usman Shaukat, whose area of research is on multi-step attack detection using event log correlation.

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Disclosure

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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